Chapter test

43 Bioethics

Name: _____ ( ) Class: _____ Date: _____

Total marks: 30 marks
Time allowed: 30 minutes
Answer ALL questions.

Score: / 30

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is ***not*** a concern about GM food?

   A. GM food may cause allergy.
   B. GM food can be produced in higher yields and at lower production costs.
   C. The long-term effects of consuming GM food on health are unknown.
   D. Vegetarians may unknowingly consume food that contains animal genes. ☐

2. Which of the following is ***not*** an environmental issue raised by biotechnology?

   A. Farmers only concentrate on planting profitable GM crops and thus biodiversity is adversely affected.
   B. GM crops are better adapted to the environment and thus may replace their wild types.
   C. GM crops may compete with the other plants in the same habitat.
   D. Planting insect-resistant GM crops may cause less chemical pollution. ☐

3. Which of the following is ***not*** an advantage of growing GM crops?

   A. The nutritional value of GM crops is higher.
   B. The shelf life of GM crops is longer.
   C. GM crops are more resistant to pests.
   D. GM crops can out-compete their natural counterparts. ☐

4. Which of the following activities may cause ethical concerns?

   (1) cloning humans to obtain organs for transplant
   (2) manipulating embryos to create ‘tailor-made baby’
   (3) using transgenic animals for drug testing

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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5. Which of the following statements about cloning are correct?

   (1) Endangered species can be conserved by cloning.
   (2) Extensive cloning of endangered animals can provide more genetic variations for evolution.
   (3) The success rate of cloning is low and many embryos are sacrificed in experiments.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

6. Which of the following statements about gene therapy is correct?

   A. The cost of gene therapy can be afforded by the general public.
   B. Gene therapy brings concerns about the right to modify one's genetic make-up.
   C. Gene therapy is banned in all countries.
   D. Gene therapy is proved to have no long-term side effects on patients. ☐

7. Which of the following is ***not*** a reason against somatic cell gene therapy in treating genetic diseases?

   A. Repeated treatments are needed regularly.
   B. The introduced gene will not be passed onto the offspring.
   C. The process of natural selection will be interfered by humans.
   D. The genetic make-up of the patient's offspring is modified. ☐

8. Which of the following is ***not*** a dispute over genetic testing?

   A. People can choose whether or not to receive genetic testing for disease-related genes.
   B. People may live with distress if they are diagnosed of carrying disease-related genes.
   C. Parents can screen the genetic makeup of embryos and abort the embryos with genetic defects.
   D. Disclosure of personal genetic information may cause discrimination against the people with genetic abnormalities. ☐

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9. Which of the following descriptions about stem cell transplant are correct?

(1) The operation cost is very high and cannot benefit the general public.
(2) The transplanted stem cells will not be rejected by the patient's body.
(3) The use of human embryos has caused controversy.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

10. Which of the following is ***not*** a possible concern about cloning?

A. Fast propagation of cloned species is achieved.
B. Biodiversity is adversely affected.
C. Scientists only propagate extensively the organisms with desired traits.
D. Humans may be cloned for the desired characters. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	43.2	43.2	43.2	43.5	43.3	43.5	43.5	43.4	43.6	43.3

B. Problem solving (9 marks)

★ 11. Papaya is a tropical fruit that is susceptible to Papaya Ringspot Virus (PRSV). A transgenic papaya plant resistant to PRSV has been produced with the coat protein gene of PRSV inserted. The transgenic papaya plant synthesizes the coat protein of PRSV in it cells.

(a) Some people believe that eating the transgenic papaya described above might cause an allergic reaction that the non-GM papaya does not. Explain why this claim is reasonable.
(2 marks)

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(b) Some people worry that the cultivation of transgenic papaya might contaminate the farmlands where non-GM papayas are grown. Suggest how this might happen. (2 marks)

(c) If GM crops are released to the natural environment, they may cause the wild strains of the same species to become extinct. Explain why the wild types might become extinct. (3 marks)

(d) Suggest a way in which the following parties can do respectively to minimize the chance of releasing GM crops into nature.

(i) farmers (1 mark)

(ii) the Government (1 mark)

(Total: 9 marks)

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C. Exam challenge (11 marks)

★★ 12. Omega-3 fatty acids are good for the health of the cardiovascular system. They are found at high concentrations in oily fish and flax seeds. Omega-6 fatty acids are another type of lipids that are common in meat. Scientists are studying the use of selective breeding or genetic engineering to produce pork from pigs which contains a higher concentration of omega-3 fatty acids.

(a) Peter suggests that feeding pigs with oily fish could help to raise the pig's omega-3 fatty acid content. Explain why this method may not work. (2 marks)

(b) Suggest how selective breeding could be used to produce pigs which contain a higher concentration of omega-3 fatty acids. (2 marks)

(c) Scientists target a gene from a worm to be inserted into pigs. This gene codes for enzyme X which catalyses the conversion of omega-6 fatty acids into omega-3 fatty acids.

(i) Suppose you are provided with copies of the gene coding for enzyme X, outline how genetically modified pig embryos could be produced using microinjection. (3 marks)

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(ii) Scientists produced 1600 genetically modified pig embryos, but only 6 piglets were born containing the worm gene. Half of them died of heart failure within a few days. Cells from one of the surviving healthy piglets were used to clone 8 more piglets. Meat from the cloned piglets contained 3 times the concentration of omega-3 fatty acids than the meat from normal pigs.

Discuss the reasons for supporting and opposing the above study. (4 marks)

(Total: 11 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	B	2.	D	3.	D	4.	D	5.	B
6.	B	7.	D	8.	A	9.	B	10.	A

(1 mark each)

B. Problem solving (9 marks)

11. (a) The transgenic papaya plant synthesizes the coat protein of PRSV in it cells. This protein is not found in non-GM papaya. (1)
The human body may recognize the coat protein of PRSV as foreign and initiate immune responses to eliminate the protein. (1)

(b) The pollen grains of transgenic papaya are carried to the wild types by pollinators. (1)
The seeds of transgenic papaya may disperse to other farmlands or grow in the wild. (1)
(Or other reasonable answers)

(c) GM crops carry modified genes which enable them to be better adapted to the environment. (1)
The GM crops compete with the wild types in the same habitat for water and nutrients. (1)
The GM crops may out-compete and eventually replace the wild types. (1)

(d) (i) Plant GM crops in isolated farmland / greenhouse. (Or other reasonable answers) (1)
(ii) Enacting laws regulating the growth of GM crops. (Or other reasonable answers) (1)

(Total: 9 marks)

C. Exam challenge (11 marks)

12 (a) The ingested omega-3 fatty acids will be metabolized by the body. (1)
Some may be converted to be used in respiration. Not all of them are used to build muscles. (1)

(b) Variation in omega-3 fatty acid content exists among pigs. (1)
Breed pigs which contain more omega-3 fatty acids. (1)
Repeat in subsequent generations. (1) (Max. 2)

(c) (i) Egg cells and sperms are obtained from pigs and are fertilized *in vitro*. (1)
The gene coding for enzyme X is injected into (the pronuclei of) the fertilized eggs. (1)
The fertilized eggs containing the gene coding for enzyme X are cultured and developed into embryos. (1)

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(ii) Supporting: (Max. 2)

- Potential benefit for human health (1)
- Successful production of more genetically modified piglets using cloning with high success rate (1)

(Or other reasonable answers)

Opposing: (Max. 2)

- Many embryos die or are discarded. (1)
- Offspring may suffer from poor health, e.g. some piglets have heart failure. (1)
- Introduction of new genes has unforeseen consequences for future generations of pigs. (1)
- Ingestion of meat containing foreign genes may cause allergies or other unknown health effects on people. (1)

(Or other reasonable answers)

(Total: 11 marks)

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